

Detecting Tennessee Eastman faults from normal operation only

Miniproject. PCA retained $k = 31$ components (90.2% of the training variance). Thresholds are the empirical 99th percentile of validation runs 301-400.

1. The plant and the task

The Tennessee Eastman Process is a simulated chemical plant. Gaseous reactants enter an exothermic reactor, the product is condensed and recycled, and a stripper sends the liquid product downstream. Fifty-two signals are recorded every three minutes: forty-one measurements and eleven valve positions. A run is 500 samples, which is 25 hours. In a faulty run the disturbance starts at one hour, so samples 1 to 20 are still normal and samples 21 to 500 are not.

Faults are uncommon, and the next one is often a kind nobody has labelled. A classifier trained on old fault examples cannot see that new kind. Process monitoring therefore learns what normal operation looks like, and raises an alarm when a new sample stops looking normal. This project builds two such detectors. Both are trained only on fault-free runs 1 to 300. Thresholds use runs 301 to 400. False alarms use runs 401 to 500. The same twenty faults are then scored. No faulty row is used to fit a model or to choose a threshold.

2. The two detectors

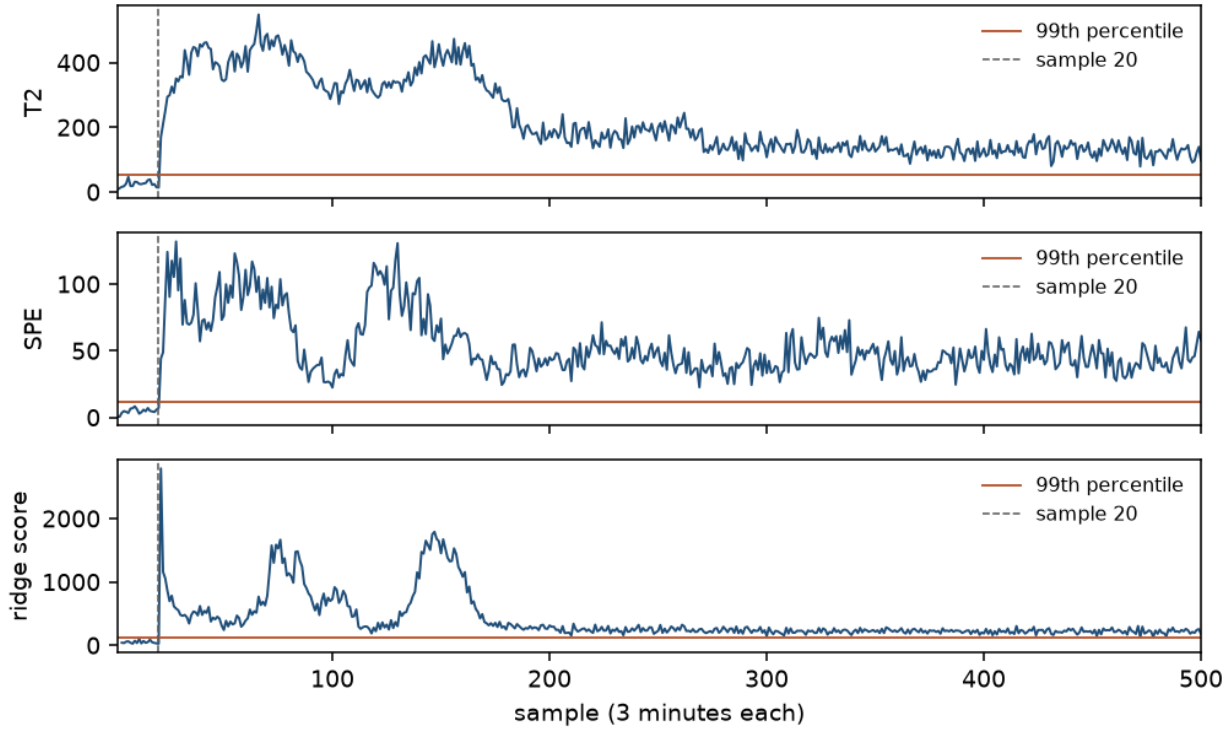
The PCA monitor standardizes each channel by the training mean and the training standard deviation, using divisor $n - 1$, then keeps the smallest set of principal components whose eigenvalues sum to at least 90 percent of the total. That cutoff is $k = 31$ (90.2%). For a standardized row z and loadings P , the score vector is $t = P^T z$. T^2 sums t_i^2 / λ_i for the retained components. It is large when the plant still moves along its usual directions, but by an unusual amount. SPE is the squared length of the residual $z - P P^T z$. It is large when the channels stop moving together, so the retained components cannot reconstruct the row.

Project primer: "T2 is the squared distance from the centre in that subspace, with each direction scaled by its own variance." "A large SPE means the channels have stopped moving together the way they normally do."

The forecast monitor is the Lecture 8 residual detector applied to all 52 channels. Inside each run the features are the standardized rows at $t-1$ and at $t-2$, 104 columns, and the target is the standardized row at t . One ridge regression with $\alpha = 1$ is fit on the training runs and predicts all 52 channels at once. Each residual is divided by that channel's training residual standard deviation, again with divisor $n - 1$, then squared and summed. The first two samples of a run have no score, because $t-2$ does not exist, and a lag never crosses a run boundary. This score is large when the recent past does not predict the present. That is a different question from asking whether the operating point is far from the normal cloud.

Each threshold is `numpy.quantile` of the validation scores at probability 0.99, with NumPy's default interpolation. T^2 is 51.8, SPE is 11.77, and the ridge score is 112.4. A sample is in alarm only when it and the two samples before it all exceed that line. Figure 1 is fault 7, run 1 (C header pressure loss, reduced availability (stream 4); step). On that run, the share of samples above the line in samples 21-40, then in samples 200-500, is T^2 1.00 then 1.00, SPE 1.00 then 1.00, and ridge 1.00 then 1.00. The dashed line is the last normal sample. The horizontal line is the validation 99th percentile of that statistic.

Fault 7, run 1: C header pressure loss, reduced availability (stream 4); step



3. Results

The false-alarm rate is the share of samples in alarm on fault-free test runs 401 to 500. T2 0.0005, SPE 0.0000, ridge 0.0000. The detection rate, for each faulty run, is the share of samples after sample 20 that are in alarm, averaged over the 20 runs. Delay is the median, over the runs that did alarm, of the minutes from sample 20 to the first later alarm. Samples are 3 minutes apart, so an alarm at sample 21 is a delay of 3 minutes. The cell "none" means every run was missed. A count after the delay is how many of the 20 runs never alarmed.

Fault	T2 rate	T2 delay	SPE rate	SPE delay	Ridge rate	Ridge delay
1	0.985	25.5 min	0.992	15 min	0.994	12 min
2	0.973	39 min	0.959	57 min	0.965	51 min
3	0.000	231 min, 19 missed	0.000	none	0.000	none
4	0.091	64.5 min	0.994	12 min	0.969	9 min
5	0.332	9 min	0.125	15 min	0.996	9 min
6	0.979	33 min	0.996	9 min	0.996	9 min
7	0.996	9 min	0.996	9 min	0.996	9 min
8	0.942	70.5 min	0.822	67.5 min	0.954	52.5 min
9	0.000	231 min, 19 missed	0.000	none	0.000	none
10	0.085	471 min	0.177	177 min	0.698	67.5 min
11	0.258	43.5 min	0.434	40.5 min	0.454	33 min
12	0.944	37.5 min	0.867	48 min	0.957	30 min
13	0.884	159 min	0.893	136.5 min	0.902	138 min
14	0.978	12 min	0.784	13.5 min	0.995	9 min
15	0.000	231 min, 19 missed	0.000	none	0.000	none
16	0.019	690 min, 3 missed	0.101	156 min	0.712	45 min
17	0.666	138 min	0.868	118.5 min	0.867	118.5 min
18	0.871	175.5 min	0.884	151.5 min	0.893	141 min
19	0.010	471 min	0.003	610.5 min, 6 missed	0.029	121.5 min
20	0.136	186 min	0.432	165 min	0.635	151.5 min

4. Comparison

T2 is large inside the normal subspace, and a quiet direction counts more because its eigenvalue is in the denominator. SPE is large when channels stop moving together. The ridge score is large when the last two samples fail to predict the present. A rate of at least 0.50 is a clear catch; a rate below 0.10 is a miss. On that 0.50 line, every fault SPE catches, ridge catches too. Ridge also clearly catches 5 (condenser cooling water inlet temperature), 10 (C feed temperature (stream 4)), 16 (unknown), 20 (unknown). SPE clearly catches fault 4 and T2 does not: T2 0.091, SPE 0.994, ridge 0.969. No fault is clearly caught by T2 and missed by SPE. The faults below 0.10 on every statistic are 3 (D feed temperature (stream 2)), 9 (D feed temperature (stream 2)), 15 (condenser cooling

water valve), 19 (unknown).

Fault 4 is a step in reactor cooling-water inlet temperature. On run 1, the share of samples above the line in samples 21-40, then in samples 200-500, is T2 0.15 then 0.35, SPE 0.90 then 1.00, ridge 1.00 then 0.99. The temperature loop moves the cooling-water valve and pulls the measurements back toward the normal cloud, so T2 does not stay in alarm (rate 0.091, though every run does alarm at least once). SPE stays large because that valve and the reactor temperature are no longer on their normal relationship. The forecast was trained on the normal relationship, so its residual stays large as well.

Fault 5 is a step in condenser cooling-water inlet temperature, and it is the case where the detectors disagree. Just after the fault on run 1, all three are over the line (T2 0.90, SPE 0.90, ridge 1.00). Late in that run T2 is 0.03 and SPE is 0.04, while ridge is still 1.00. The controllers bring the static operating point back inside both PCA limits. The one-step forecast does not recover. Averaged over the 20 runs the rates are T2 0.332, SPE 0.125, ridge 0.996. Fault 7 in Figure 1 is the opposite pattern: the disturbance is not absorbed, and the late shares stay high (T2 1.00, SPE 1.00, ridge 1.00).

5. Faults nobody catches

Faults missed by all three statistics, under that same 0.10 line: 3 (D feed temperature (stream 2)), 9 (D feed temperature (stream 2)), 15 (condenser cooling water valve), 19 (unknown). The project check, and the Tennessee Eastman studies it points to, single out faults 3, 9, and 15. Fault 3 is a step in D-feed temperature, fault 9 is random variation of that same temperature, and fault 15 is a sticking condenser cooling-water valve. Monitoring papers call these three unobservable: on the 52 recorded channels they look like normal operation. The temperature disturbance barely appears in the measurements, and the level and temperature loops can absorb the sticking valve. Our rates are fault 3: T2 0.000, SPE 0.000, ridge 0.000; fault 9: T2 0.000, SPE 0.000, ridge 0.000; fault 15: T2 0.000, SPE 0.000, ridge 0.000. Runs missed, written as T2 / SPE / ridge, are fault 3: 19/20/20, fault 9: 19/20/20, fault 15: 19/20/20.

That says something about these sensors and this controller, not that the plant is healthy. A disturbance can be real and still be invisible when it is rejected before it reaches a recorded channel, or when it only rearranges the plant inside the variation already present in the 300 normal runs.

6. Diagnosis

SPE and the ridge score are sums of squares, so each splits into one term per channel. For each fault those terms are averaged over every sample that is in alarm after sample 20, across the 20 runs, and the top five channels are kept. The average shows where the fault appears. Westerhuis, Gurden and Smilde (2000) show that contribution plots smear: a channel correlated with the disturbed one inherits part of the score, so the largest term is not always the place the fault started.

Fault 1: A/C feed ratio, B composition constant (stream 4); step. SPE top five: xmeas_4 (A and C feed (stream 4)); xmv_4 (A and C feed flow valve (stream 4)); xmeas_38 (product composition E); xmeas_21 (reactor cooling water outlet temperature); xmeas_33 (purge composition E). Ridge top five: xmeas_1 (A feed (stream 1)); xmv_3 (A feed flow valve (stream 1)); xmeas_16 (stripper pressure); xmeas_19 (stripper steam flow); xmeas_7 (reactor pressure). The fault is an A/C feed-ratio step in stream 4. SPE's two largest terms are the stream-4 flow and its valve, which is where the fault is introduced. Ridge instead puts almost all of its score on the A feed and the A-feed valve: the ratio change shows up there as a correction. Product composition E and the reactor cooling-water outlet temperature in the SPE list are downstream of that feed change. The detectors agree that a feed moved, and they disagree which leg is loudest.

Fault 4: reactor cooling water inlet temperature; step. SPE top five: xmv_10 (reactor cooling water flow valve); xmeas_9 (reactor temperature); xmeas_33 (purge composition E); xmeas_25 (reactor feed composition C); xmeas_29 (purge composition A). Ridge top five: xmv_10 (reactor cooling water flow valve); xmeas_9 (reactor temperature); xmv_3 (A feed flow valve (stream 1)); xmeas_1 (A feed (stream 1)); xmeas_26 (reactor feed composition D). The inlet temperature itself is not one of the 52 channels. Both detectors rank the reactor cooling-water valve first and the reactor temperature second, which is the loop that answers the disturbance. The other SPE terms are much smaller, so this ranking is not smeared across the plant.

Fault 6: A feed loss (stream 1); step. SPE top five: xmv_5 (compressor recycle valve); xmv_10 (reactor cooling water flow valve); xmeas_9 (reactor temperature); xmeas_20 (compressor work); xmeas_1 (A feed (stream 1)). Ridge top five: xmeas_1 (A feed (stream 1)); xmv_3 (A feed flow valve (stream 1)); xmeas_10 (purge rate (stream 9)); xmv_6 (purge valve (stream 9)); xmv_5 (compressor recycle valve). Ridge ranks the A feed and its valve first, which is the stream that was lost. SPE's largest term is the compressor recycle valve, then reactor cooling and compressor work; the A feed itself is only fifth. The plot is showing the plant-wide correction, not only the stream that failed. That is the smearing Westerhuis, Gurden and Smilde describe, and it is also a real material balance: losing the A feed forces the recycle and the reactor heat balance to move.

7. Limits

The threshold is an empirical percentile of held-out normal runs, not the textbook F limit for T2 or the Jackson-Mudholkar approximation for SPE. Those limits assume independent normal samples. Samples taken three minutes apart are neither, which is why the recipe uses one percentile for all three statistics. The number means that about one percent of normal validation rows sit above the line. The three-in-a-row rule then makes the test-set false-alarm rate smaller, which is what the fault-0 row reports.

This setup cannot say what a real plant would do. The records are one simulated operating mode with scripted faults. They do not include a slow drift of the normal regime, a sensor that fails stuck, or a fault that arrives during a setpoint change. Contribution ranks are not root causes. The ridge model is linear and looks back only two samples, so a slow oscillation can look predictable. A fault absorbed by the controllers stays invisible to both detectors, however the threshold is set.

8. AI use

The pipeline and this report were drafted with Cursor's coding agent (Grok). The draft was required to follow the project recipe: training runs 1-300, validation runs 301-400 for thresholds only, test runs 401-500 for false alarms, standardization with divisor $n - 1$, a 90 percent PCA cutoff, Ridge with alpha 1 on lags $t-1$ and $t-2$, `numpy.quantile` at 0.99, the three-sample alarm rule, and the top five contributions on alarming samples after sample 20. The course evidence script was not run. The formulas and the filters were checked in the code against that recipe, and the table in section 3 is computed from the score files rather than typed in. The PDF was not edited by hand after it was written. Each author still has to be able to explain the code they are responsible for and the comparison in section 4.

Appendix: contributions

This appendix does not count toward the four-page limit. One entry per member.

Sean (ANDREW_ID, schao1215@gmail.com). Shared preprocessing, PCA monitor, ridge forecast-residual monitor, thresholds, alarm rule, detection table, contribution table, figures, and this report. The pipeline was drafted with Cursor's coding agent and checked against the project recipe.

The team has four members. Add the other three Andrew IDs, and what each person built, before this file is submitted. The evidence script does not read this page.